Sahaj Bhadja

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Summary

Software Engineer with an MS in CS and 2+ years shipping full-stack features in **TypeScript**, **Node.js**, React, and Python, including a home-services **booking** platform with real-time scheduling and production **AI** workflows, built daily with Claude Code.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Web Technologies, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Rebuilt production **AI workflows** as fault-tolerant FastAPI background jobs with explicit state transitions and failure isolation, processing 1K+ LLM jobs at 3x throughput without cascading failures under concurrent load.
- Debugged slow dashboards on a multi-tenant platform for 500+ users, redesigning indexes and access patterns to cut latency from 1.2s to 350ms, while integrations held 99%+ production **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Owned an **AI product** end to end, from React views to Flask APIs and multimodal LLM grading with GPT-4o, Claude, and Qwen-VL, serving 150+ instructors and cutting feedback turnaround 60%.
- Handled real-world **edge cases** like unreadable handwriting and provider timeouts with OCR fallback routing, keeping 10K+ daily requests stable at peak load and processing 1K+ uploaded files per batch.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated cross-platform data-sharing **workflows** that cut manual data entry 35%, working directly with Microsoft and GitHub stakeholders from requirement analysis through deployment and production support in Agile sprints.
- Wrote 20+ unit and integration **test** suites in nightly Jenkins pipelines and debugged large-scale API integrations with caching and pagination, improving response times 30% and cutting technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **TypeScript** and **Node.js** microservices with React for resource allocation, customer records, billing, leads, and SMS and email notifications, syncing 2M+ records across multiple locations in real time.
- Built a **resource allocation** service with real-time assignment tracking and conflict resolution, and used analytics to drive changes that cut resource wastage 42% and raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built backend APIs for a home-services **booking** app covering cleaning, AC repair, and cooking, with time-slot selection, coupons, and Firebase real-time status updates, sustaining sub-200ms latency on core booking flows.
- Prevented double-booked time slots with Firebase transactional locking and automated refunds through idempotent **webhooks**, speeding booking confirmations 40% and cutting manual payment handling 70% across 50K+ bookings.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, C++
Full Stack: Node.js, Express.js, React, Next.js, React Native, FastAPI, Flask, REST APIs, WebSockets, Webhooks
AI: Claude, Claude Code, GPT-4o, LLM Workflows, Prompt Chaining, OCR, RAG, Twilio
Data & Cloud: PostgreSQL, MongoDB, Firebase, MySQL, AWS, Azure, Docker, CI/CD
Fundamentals: Data Structures, Algorithms, Debugging, Unit Testing, Integration Testing, Code Reviews
Practices: Customer Discovery, Requirements Gathering, Agile

Projects

AI Job Booking and Dispatch Assistant | *TypeScript, Node.js, Twilio, Claude, PostgreSQL, React*

- Designed a TypeScript and Node.js voice assistant that answers **Twilio** calls, qualifies service requests with Claude, books open slots, and **dispatches** technicians by skill, distance, and availability.

HMIS – Hospital Management Information System Live | *TypeScript, MERN, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript **scheduling** app for appointment booking, rescheduling, and cancellation with Twilio SMS reminders and confirmations, managing 100+ daily appointments and cutting admin workload 30%.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented six **scheduling algorithms** including FCFS, SJF, Round Robin, and multilevel queues in C++ with a React simulator, testing 1K+ process combinations and showing 25% lower average turnaround.